

REVIEW

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Colours within planetary boundaries? Applying the Three Horizons framework to a critical review of natural dyes in the textile industry sustainability transformation

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Abstract

As a fossil-dependent industrial system, the fashion and textile industry is operating outside safe planetary boundaries and requires a systemic sustainability transformation. Starting from this premise, this review focuses on natural textile colourants with two complementing aims. First, to examine how the present and future of natural colourants are discussed and framed in the research literature. Second, to explore how these framings situate natural colourants in the larger context of textile industry sustainability transformation. The paper synthesises the literature on natural textile colourants in socio-cultural contexts beyond the laboratory, combining reflexive thematic analysis as a method and the Three Horizons framework of change as an analytical tool. The literature reviewed portrays natural colourants as a re-emerging field with potential to challenge and reshape established textile practices and thus support sustainability transformation in the industry. Simultaneously, natural colourants are burdened by the dominance of synthetic colourants as well as the established consumer and producer mindsets concerning, for example, the abundance, appearance and cost of products. To balance with these tensions requires the fashion and textile industry to consider both (infra)structural and mindset lock-ins, fragmented system perspectives and the unavoidable trade-offs of change. By applying the Three Horizons framework to the analysis of literature, this review carries out a futures-oriented reading of natural colourants, drawing attention to the performative and instrumental roles of our images of the future and a transformative intent in making comprehensive change. Whichever trajectory the textile industry considers worth pursuing will have resources devoted to it, improving it and giving it an advantage over any other alternative future.

Keywords Sustainability transformation, Sustainable textile, Natural dye, Bio-based colourant, Textile dyeing, Images of the future

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Introduction

This review adopts as its premise that the fashion and textile industry, as a heavily fossil-dependent, material intensive and global industrial system, is increasingly constrained by and operating outside of safe planetary boundaries and requires a systemic sustainability transformation [58, 76]. Against this backdrop, the review focuses on natural textile colourants as part of that transformation. At the intersection of cultural heritage, technological innovations, materiality research and environmental science, natural colourants offer an illustrative case and a material lens through which to discuss established and largely fossil-dependent industrial systems' socio-technical change towards more regenerative and biobased models.

The question of natural colourants is situated within a broader discourse on fossil fuel phase-out in our societies. In the textile context this means finding alternatives to synthetic fibres and fossil-fuel-powered production and transportation, but also the dominant, petroleum-derived synthetic dyes and pigments. Despite the constant presence of coloured objects in our lives, this materiality of colours has often been overlooked – colour is the visible and foundational part of design and products yet largely hidden from the discussions about sustainable consumption and production futures. Dyeing processes significantly contribute to the water and carbon footprints of textiles, and chemicals have impacts on the health of individuals and ecosystems. Globally, textiles are the largest users for synthetic dyes (soluble), while paints and coatings are the primary users for pigments (insoluble) (e.g., [53]). Increasingly since the 2010s, natural colourants have gained public and research interest [52], as the awareness of the risks posed by synthetic dyes and an intention to move away from fossil-fuel-based substances have motivated the search for alternative ways to colour our goods.

There is broad consensus on the *why* of sustainability transformations, but less agreement on the *how*: the *means* (e.g., actions, solutions, tools), and the *manners* (e.g., values, visions, principles) [8]. In this paper we explore these means and manners of textile industry sustainability transformation through the role of natural colourants, with two complementing aims. First, to examine how the present and alternative futures of natural colourants are articulated and framed in the literature. Second, to explore how these framings situate natural colourants in the larger context of textile industry sustainability transformation – whether natural colourants are constrained by the current textile industry conventions or seen as transformation drivers. These explorations are guided by the Three Horizons framework, discussed in the next section. Through the framework's distinct time perspectives, we look at how the

dominant textile colouration practices are problematized (present Horizon-1), what alternative futures for natural colourants can be identified (future Horizon-3), and if and how transformational intent towards natural colourants is portrayed in the literature (transformational middle Horizon-2). Through these questions this review aims to contribute to “the rich tapestry of understanding” [11], p. 120) of natural colourants' positioning and prospects in the industry.

Research on natural colourants often focuses on dye performance and technologies [88], taking place in experimental laboratory settings, where dyestuffs and auxiliary products are isolated and analysed, and fibres and fabrics are dyed and tested. To expand the perspectives, this review has gathered studies that have been conducted beyond the laboratory and examine natural colourants in applied, practice-based contexts, such as markets, consumer behaviour, brand practices, and policymaking. We focused on the dynamics of sustainability transformation and the transformative potential of images of the future, perspectives which have not previously been highlighted in the context of natural colourants. Rather than discussing the prospects of or building scenarios for specific technologies or colourant sources, we explored how time horizons, mindsets, and values can drive or restrain change. Despite its scale, complexity and global impacts, the fashion and textile industry has received relatively limited attention within futures research. With our novel research approach, we aim to contribute towards narrowing that gap. We hope the paper is of particular interest to those fashion and textile professionals working in research, industry and/or decision-making, who are invested in sustainable textile futures and wish to challenge the dominant logics of the current textile sector. The paper encourages having critical conversations about change and continuity in the many situations where the futures of textiles are discussed and made.

For brevity and for consistency with the predominant terminology of the literature reviewed, we have used the term natural colourant to refer to both dye sources extracted from nature and to more processed dyes derived from natural origins, a broader category of biobased colourants. The colourant sources discussed in the literature include plants and fungi, microbes, minerals, insects, and by-products from agriculture, forestry, and the food and beverage industries.

In the following sections we first discuss the Three Horizons framework and its application in a textile colouration and sustainability transformation context. Then we describe the research design, including data collection and analysis method reflexive thematic analysis. The next section discusses findings and their implications, followed by reflections on the study and its limitations as well as future research ideas. The article

concludes by synthesising the findings and situating the analysis within broader discussions on transformations and temporalities.

Sustainability transformation and the Three Horizons framework

In this paper, we understand sustainability transformation as a shift towards a society that operates inside the planetary boundaries. Seven of these nine essential environmental thresholds are assessed to have passed their safe limits (see [64, 70, 71]). All the crossed planetary boundaries are directly influenced by the textile industry through land, energy and freshwater use, chemical load, and pollution. Within these planetary limits, sustainability transformation should also ensure socially just well-being (see [69]). We use *transformation* to describe a disruptive, comprehensive change, typically including several smaller, often more technical *transitions* in our industrial and everyday practices. For discussions on the nuances of terminology see, e.g., Child and Breyer [19] and Hölscher et al. [36].

Here, the Three Horizons framework (Fig. 1) provides an analytical heuristic for exploring sustainability transformation dynamics over time. The framework conceptualises the future as interaction between three temporal horizons – the dominant present, emergent future, and a disruptive, transitional middle [79], p. 10). Sharpe et al. [80] propose that the Three Horizons provides a simple structure for working with complexity and helps develop an awareness of the future and its potential in the present.

They also argue that the framework supports finding ways to manage transitions and distinguish between incremental and transformative change. Additionally, it may help make power structures explicit and encourage dialogue between actors with different mindsets. Despite the framework being a simplification of a complex and an uncertain reality, these proposed benefits suggest the Three Horizons to be a relevant tool for our research setting, where the objective is to understand natural colourants as part of wider textile system transformation dynamics and beyond the present. While the framework is commonly used to facilitate dialogue at participatory or strategic contexts, it has also been applied analytically (e.g., [41]). We adopted it as a tool to analyse data that was originally not created using the framework's guidance, complementing the framework's participatory use and exploring its potential to support reflexive analysis of existing literature.

The horizons are typically approached in a non-chronological order of 1–3–2, to first consider the present (1), then imagine the future (3), and lastly look at the transformative in-between period (2). The first horizon (Horizon-1) illustrates the current way of doing things, the dominant infrastructures and ways to organise our societies. We are dependent on the services Horizon-1 provides, and it may have served us well until now [80]. In our setting, synthetic colourants represent Horizon-1. They are the prevailing means for adding colour to textiles, dominant since their invention in the 1850s [84], p. 9). Synthetic

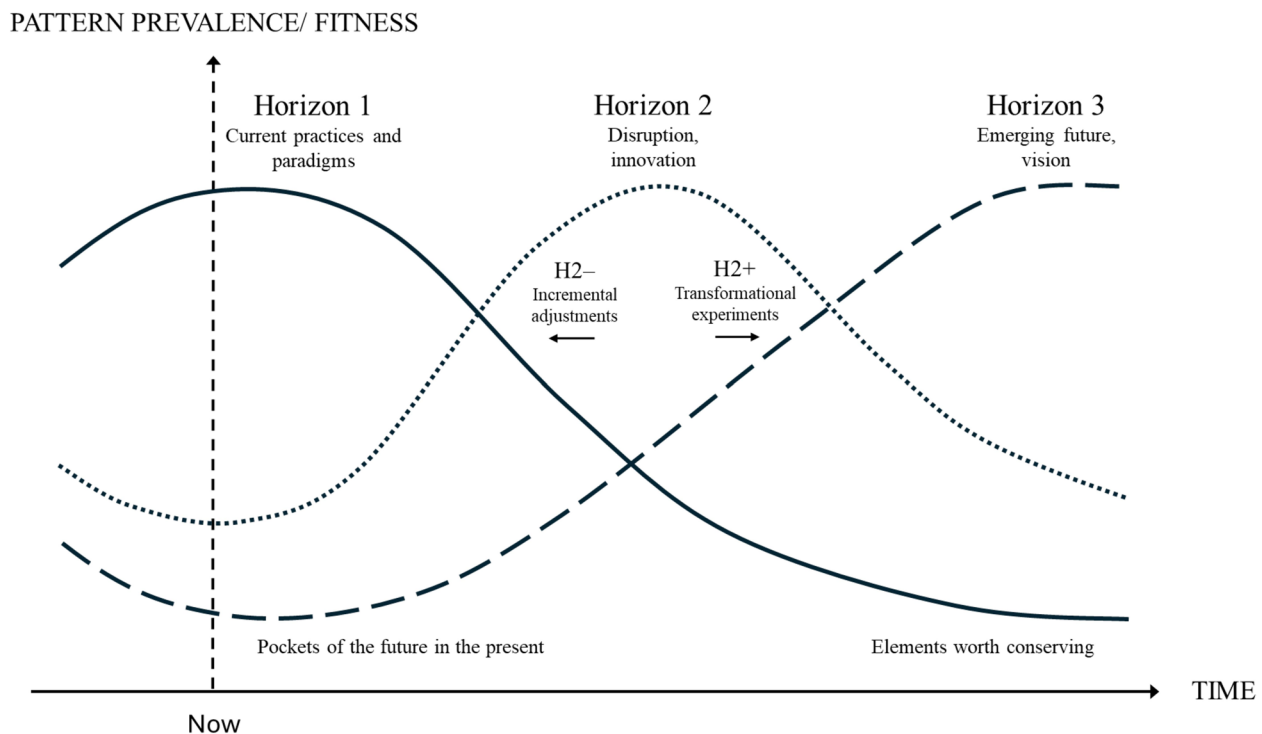


Fig. 1 The Three Horizons framework, adapted from Fazey and Leicester [31] and Sharpe et al. [80]

colourants enabled the production and use of a near-endless range of inexpensive, stable, and reproducible colours, quickly and almost entirely replacing the use of natural colourants. However, with the growing volume of dye demand and accumulated knowledge of drawbacks and risks of petroleum-based substances, Horizon-1 is losing its fitness for purpose. Still, Sharpe et al. [80] note that an important step in the Three Horizon practice is to recognize that each horizon also has valuable elements or essential features worth preserving. What will be let go of needs to be done with care and control, as systems typically cannot be shut down before alternatives exist. The figurative Horizon-1 mindset or voice is that of a manager, someone responsible for keeping things going, thus typically unwilling to take risks or make major changes [80].

The third horizon (Horizon-3) represents the future, a potentially better alternative to Horizon-1. At any moment there are multiple competing third horizons [21], the triumph of natural colourants, hypothesised here, only one of these alternative futures. Similar to weak signals, signs of Horizon-3 can often be found as “pockets of the future in the present” [21]. These natural colourant pockets are everywhere in the world, possibly gaining momentum in artisans’ workshops, museums, research laboratories, and local businesses. The metaphorical Horizon-3 mindset is a visionary one, innovative yet sometimes idealistic [80].

The second horizon (Horizon-2) is depicted as a transitional period between the present and the future. It is often turbulent, with old and new ideas and practices colliding. Despite the transformative premise of Horizon-2, innovations are often biased to the past and present, as most innovation resources are in the hands of the Horizon-1 providers. It is also easier to innovate within existing systems than to challenge them. Therefore, some Horizon-2 initiatives are just prolonging Horizon-1, keeping it going a bit longer, or being absorbed by it (H2–). Other initiatives are truly making a change and taking us towards something new (H2+) [79], pp. 21–22, [80]. Respectively, despite being an alternative way to add colour to a substrate, some ideas about natural colourants may still focus more on how to make them fit inside the frames of the current industry logics and practices (H2–), instead of challenging them (H2+). The symbolic mindset of Horizon-2 is entrepreneurial, someone eager to try new things, with varying degrees of success [80].

The Three Horizons framework is visualised in Fig. 1. With time (x-axis), the horizons fluctuate, losing and gaining fitness for purpose and strength (y-axis).

Materials and methods

A critical-integrative review approach was chosen for its suitability to our research setting: an interest in emerging issues and working with conceptual insights.

Integrative reviews critique and synthesise literature, creating new frameworks and perspectives on the topic, resolving inconsistencies in the literature and catalysing future research [29, 83]. Within this critical-integrative approach, the Three Horizons framework provided an analytical focus on temporal dynamics in the data, and reflexive thematic analysis [10, 11] was applied to navigate the analysis process.

Data collection

The data collection was conducted systematically. Four databases were searched: Scopus, Web of Science, Academic Search Complete, and ProQuest (database Art, Design & Architecture). An additional search was carried out by using the Google Scholar search engine, and the first 500 search results were screened. The snowball method was also used to hand search references cited in studies already identified. The search was limited to peer-reviewed journal articles, book chapters and conference papers published in English in the 15-year period between 2010 and 2024. The search was adjusted according to minor differences in the search logics of the databases. For the core search query, see Supplementary Fig. 1.

Inclusion, exclusion and quality criteria

Publications meeting the search criteria were included if they discussed natural colourants in textile applications and in industrial and/or socio-cultural contexts, outside purely experimental laboratory settings. Review articles were excluded, except for a few synthesising book chapters. The search flow with numbers of identified, excluded, and included studies is presented in Supplementary Fig. 1. The studies reviewed ($n = 41$) are listed in Table 1.

Studies with a country-specific dataset provided a degree of geographical heterogeneity: 13 different countries were represented, mostly from Asia, Europe, and North America as well as one dataset from South Africa. The literature covered several focus areas, including consumer perceptions and preferences, dye producer and end-product producer perspectives, design and aesthetics, green marketing, natural colourants as tradition and heritage, collaboration, bio-colourant database and standards, and the questions concerning craft-scale to industry-scale shift.

Thematic analysis as an analytical method

The texts selected were analysed following the process of reflexive thematic analysis [10, 11], a qualitative analysis method aiming to identify themes, or patterns, in the data. The flexible method made it possible to use both deductive and inductive approaches to analysis, and to explore the data both descriptively and interpretatively

Table 1 Overview of studies reviewed. See reference list for full details

Author(s), Year	Title	Relevance & Contribution	Publication type
Agustini MH et al. (2019) [1]	<i>Identification of green marketing strategies: Perspective of a developing country</i>	Green marketing strategies for natural dye products	Journal article
Autio M et al. (2023) [4]	<i>Consumers' concern about chemicals in clothing – Environment, skin contact, and the materiality of colour</i>	Consumers' understanding of the materiality of synthetic and bio-based colours	Conference paper
Bechtold T, Manian A (2023) [5]	<i>Technical aspects and requirements to produce natural colorants – Processes and product standardization</i>	Natural colourant standardisation	Book chapter
Bechtold T et al. (2023) [6]	<i>Textile coloration with natural dyes and pigments</i>	Natural dye applications on textiles	Book chapter
Cao H et al. (2014) [15]	<i>Locally produced textiles: Product development and evaluation of consumers' acceptance</i>	Consumer perceptions of local natural dye products	Journal article
Cardon D (2023) [16]	<i>Colour cultures: Natural dyes, a global intangible heritage with a future</i>	Natural dyes as heritage	Conference paper
Carvalho C, Santos G (2015) [17]	<i>Global communities, biotechnology and sustainable design – Natural/ bio dyes in textiles</i>	Natural dyes in the clothing industry	Conference paper
Casciani D, Chen M (2024) [18]	<i>Conveying natural dyes in the fashion industry through a design driven innovation</i>	Natural dyes in clothing design process	Conference paper
Choi J, Kim S (2013) [20]	<i>Cross-cultural study of color sensation and sensibility for Korean traditional fabrics with natural indigo dyeing</i>	Natural dye product planning for international markets; cultural differences in colour preferences	Journal article
de la Sayette A (2023) [23]	<i>Large-scale production of dye plants and plant extracts for industrial applications of natural dyes</i>	Challenges and opportunities of large-scale production of dye plants	Conference paper
Dong Z-J et al. (2023) [24]	<i>The perceptual evaluation of clothing sustainable color in clothing design</i>	Consumer colour preferences	Journal article
Doty K et al. (2024) [25]	<i>Natural dyes in the United States: An exploration of natural dye use through the lens of the circuit of style-fashion-dress</i>	Production, distribution, consumption and regulation of natural dyes in the US	Journal article
Elsahida K et al. (2019) [27]	<i>Sustainability of the use of natural dyes in the textile industry</i>	Environmental, social and economic sustainability of natural dyes	Conference paper
Elsahida K et al. (2020) [28]	<i>Sustainable production of natural textile dyes industry</i>	Sustainability of natural dyes; consumer preference and willingness to pay	Conference paper
Ferrante M. et al. (2019) [32]	<i>Socio-economic effects in the knitwear sector – A life cycle-based approach towards the definition of social indicators</i>	Life cycle assessment of natural dye sweater	Book chapter
Geissler S, Bechtold T (2023) [33]	<i>Environmental and economic position of natural colorants – Energy and resources balances, sustainability, ecology, and costs</i>	Environmental and economic perspectives of natural colourants	Book chapter
Hashim NIR et al. (2021) [34]	<i>A preliminary study of tourists and vendors perception on the development of tourism souvenirs using natural dye from plants</i>	Natural dye product development; colour tourism	Conference paper
Hong H, Kang JH (2019) [35]	<i>The impact of moral philosophy and moral intensity on purchase behavior toward sustainable textile and apparel products</i>	Natural dye product purchase behaviour	Journal article
Ingram W (2019) [38]	<i>Deities, dyers, and dollars: Balancing culture, conservation, and commerce with Indonesia's indigenous weavers</i>	Natural dyes as cultural heritage and local business	Journal article
Karadag R (2023) [42]	<i>Establishing a new international standard for natural dyed textile goods [Natural Organic Dye Standard (NODS)]</i>	Natural dye standardisation	Journal article
Kusumastuti A et al. (2023) [45]	<i>Assessment of producer's perspective on the production of environmentally friendly fashion products: A case study in Indonesian natural dyes batik craftsmen</i>	Natural dye product producers' attitude and satisfaction	Journal article
Ladekarl IM et al. (2023) [46]	<i>From craft to industry</i>	Natural dye processes potential from craft to industry scale	Journal article
Lai C-C, Chang C-E (2021) [47]	<i>A study on sustainable design for indigo dyeing color in the visual aspect of clothing</i>	Consumer colour perceptions	Journal article
Lara L et al. (2023) [48]	<i>Color in sustainable fashion: A reflection on the importance of design education</i>	Design education on sustainable colouration	Conference paper
Lin PMC et al. (2015) [49]	<i>Xinpu persimmon dye: Evolution from a local to a global industry</i>	Natural dyes as cultural heritage; colour tourism	Journal article
Mabuza L et al. (2023) [50]	<i>Natural versus synthetic dyes: Consumers' understanding of apparel coloration and their willingness to adopt sustainable alternatives</i>	Consumers' understanding and acceptance of natural dye products	Journal article

Table 1 (continued)

Author(s), Year	Title	Relevance & Contribution	Publication type
Niinimäki K et al. (2023) [59]	<i>Colour collaboration</i>	Collaboration around natural dyes within textile industry; company cases	Book chapter
Niinimäki K, Uribe LH (2023) [60]	<i>Sustainable colour aesthetics</i>	Natural dyes and acceptance of different colour aesthetics	Book chapter
Panda A et al. (2022) [62]	<i>Natural dyes – A way forward</i>	Natural dye applications on textiles	Book chapter
Patel BH (2011) [63]	<i>Natural dyes</i>	Natural dye applications on textiles	Book chapter
Primetta A, Räisänen R (2023) [66]	<i>New biocolourant database for the quality control of natural colorants and products containing them</i>	Database for natural colourants	Conference paper
Ratnapandian S (2020) [68]	<i>Natural colorants and its recent developments</i>	Natural dye applications on textiles	Book chapter
Räisänen R (2023) [75]	<i>Networking in biocolour research—creating increasing value and impact</i>	Collaboration in biocolourant research and development	Conference paper
Saxena S, Raja ASM (2014) [78]	<i>Natural dyes: Sources, chemistry, application and sustainability issues</i>	Natural dye applications on textiles	Book chapter
Uribe LH (2023) [86]	<i>Making colour a relational practice: Correspondence between materials and local environments in natural textile dyeing processes</i>	Natural colourant aesthetics; colouration practices	Book chapter
Valle-Noronha J, Puppart P (2024) [87]	<i>Estonian blues: Natural dyes as enhancers of environmental, intergenerational and heritage bonds in fashion design education</i>	Natural dyes in fashion design education; cultural heritage	Journal article
Xu H, Song H-Y (2024) [88]	<i>Key factors influencing Chinese consumers' demand for naturally dyed garments: Data analysis through KJ method and KANO model</i>	Consumer satisfaction with natural dye garments	Journal article
Xue X, Li L (2023) [89]	<i>Fashion consumption of naturally dyed products: A cross-cultural study of the consumption of blue-dyed apparel between China and Japan</i>	Natural dye fashion consumption	Journal article
Yi E, Shamey R (2015) [90]	<i>Crosscultural comparison of color terms and preference of persimmon-dyed fabric, Galchon</i>	Consumer colour preferences	Journal article
Zerin I et al. (2020) [92]	<i>Potentials of natural dyes for textile applications</i>	Natural dye applications on textiles	Book chapter
Zhou Y et al. (2012) [93]	<i>Retail demand for natural dye plants and dye plant products: A conjoint analysis</i>	Retailer interest in natural dye plants and dye plant products	Journal article

[10]. Some of the method's characteristics are considered in the section [Reflections, limitations and future research](#). ATLAS.ti software version 25.0.1.32924 [3] was used to organise the data and for the coding process.

The six phases of reflexive thematic analysis [11], pp. 35–36) guided the process and are outlined in Table 2, together with our application of the method and examples of theme development. We were interested in finding latent themes beyond explicitly stated surface meanings, interpreting underlying ideas and assumptions that were expected to shape the semantic content of the data [10], and ultimately shape the perceptions about the futures of natural colourants. This latent, or implicit, level was reached towards the end of the analysis when semantic codes were grouped and developed into themes aiming to capture more implied meanings. The level of interpretation varied across the horizons, the complex Horizon-2 content appearing to be the most layered.

Findings and discussion

The time horizons suggested by the Three Horizons framework helped identify distinct dominant, alternative and transitional dynamics in the literature. Guided by reflexive thematic analysis [10, 11], themes were created for each horizon resulting in one theme under Horizon-1,

one theme under Horizon-3, and three main themes and one integrative theme under Horizon-2.

Horizon-1 theme *Great performance, hidden harms*, captures the expressed contradictions of currently dominating synthetic colourants.

Horizon-3 theme *Growing, but not replacing*, summarises the prospects for natural colourants identified in the reviewed texts.

In the Horizon-2 perspective our analysis aimed to distinguish between initiatives that may be pulling back towards an extension of Horizon-1 (H2–), and those pushing forward towards a new Horizon-3 (H2+), resulting in three themes that highlight these tensions and transitions: *From path dependencies to a visionary attitude*; *From fragmentation and disconnection to systemic collaboration and goals*; and *From process complexity and poor performance to sustainability benefits and prerequisites*. The fourth theme in Horizon-2 is an integrative theme, *Regulation and support*, with leverage across all the other themes to discourage the use of synthetic colourants and promote natural ones. Considerations of its role are embedded into the discussion of the three main themes in Horizon-2. While Braun and Clarke [10] advise against too much overlap between themes, our Horizon-2 themes are inherently interconnected, a

Table 2 Reflexive thematic analysis process [10, 11], its application in this research, and a theme development example

Braun and Clarke's reflexive thematic analysis (2006; 2022)	PHASE 1 Familiarising with the dataset	PHASE 2 Coding	PHASE 3 Generating initial themes	PHASE 4 Developing and reviewing themes	PHASE 5 Refining, defining and naming themes	PHASE 6 Writing up
Application of the analysis process in our review	Full texts read-through, deductive data reduction guided by the research questions derived from the Three Horizons framework	1st coding round: Descriptive, inductive coding without a predefined code list [77], p. 134)	2nd coding round: Merging similar codes, assigning latent meanings [77], p. 152)	Combining groups of codes into initial themes	Finalising and naming themes	Research report written in parallel with analysis and interpretation
	Data extract <i>So far, many dye containing sources have been identified but they are simply not enough for large-scale dye production and to meet unlimited variation of color choices [92], p. 881)</i> <i>[...] we can try to investigate new versions of standards and challenge these as well as our perception of aesthetics related to colour perfection or imperfection [46], p. 4–5)</i>	Code Colour palette limitations Challenging our perceptions of perfection	Code group (H2–) Aesthetics and trends (H2+) Design and aesthetics	Candidate theme Path-dependence New aesthetics	Final theme From path dependencies to a visionary attitude	Interpretation Expectations for an unlimited variation of colours appear to project our present customs, rather than future necessities. Challenging our ideas of aesthetics and perfection was interpreted as a forward-looking mindset working with natural colourants' properties instead of against them. The final theme captures these tensions of holding on to or detaching from established ways of thinking and doing.

continuum of perspectives complementing and informing one another. Each theme is discussed in the following sections.

Horizon-1 juggles the costs and benefits of synthetic colourants

Horizon-1 theme: Great performance, hidden harms

The present textile industry logic is one of large volume, fast pace, and cost-efficiency. The literature also described it as being about stability, control, perfection, sameness, and seriality: consumers, producers and designers alike tend to consider textiles as high-quality when items produced in a batch are identical and remain unchanged over time [25, 60, 86, 87]. The current practices of short-lived fast fashion together with long-lasting synthetic colours were observed as being paradoxical by the literature reviewed [60].

Several environmental and social costs of synthetic dyes were discussed in most of the papers and expressed as arguments for the growing interests in natural dyes (e.g., [6, 48, 68, 88]). The environmental burden of synthetic colourants consists of the resource use (chemicals, energy, water), pollution from production and use (to air, soil, and aquatic ecosystems), and non-biodegradable waste generated by these processes. Social and health concerns (working conditions, carcinogenic potency) affect industry workers and may harm all textile users, directly or through environmental degradation. These costs were recognised to be largely hidden and/or externalised [33]: the harms or risks these colourants carry are not reflected in their prices nor fairly distributed. Due to their typically lower price and an established position in the industry, synthetic dyes have a competitive edge over natural ones.

As the Three Horizons framing encourages recognising what is valuable in Horizon-1, synthetic dyes were also acknowledged to have many benefits (e.g., [6, 18, 48]). They are economically inexpensive and easy to produce continuously and in large quantities. Synthetic dyes are available in thousands of shades that react predictably and repeatably, making their production and use easily managed. Their typically good light- and wash-fastness is advantageous in textiles subject to wear and tear. A durable colour is also a pro-environmental quality, potentially keeping the textile attractive and therefore in use longer. These are desirable characteristics that research on natural dyes is working on to mirror.

Horizon-3 imagines natural colourant futures

Horizon-3 theme: Growing, but not replacing

The literature recognised natural colourant “pockets of the future in the present” having persisted, or (re) appeared, in craft and laboratory scales, in niche markets, and in experimental-scale dye plant cultivation (e.g.,

[5, 46, 60]. It has been estimated that approximately 1% of textiles are dyed with natural colourants (e.g., [92]).

The studies reviewed saw future potential and market growth for natural colourants (e.g., [47, 59, 60, 63]), but more specific estimates for the growth scale were scarce. One study speculated that the proportion of natural dyes in textile production in Europe could be around 5% in an unspecified future timeframe [68]. In another study from 2023, an interviewee believed that within a decade, a “considerable” proportion of textiles will be dyed with natural colourants [59]. Despite the growth potential, several studies explicitly discussed that natural colourants will likely continue to be alternatives, not substitutes, for synthetic colourants, especially due to difficulties in scaling up natural colourant production in sustainable and economical ways [33, 68, 59].

Even with the literature being wary of natural colourants' sudden dominance, research and development were anticipated to help overcome many current challenges, and emerging technologies were expected to change the ways textiles are dyed (e.g., [48, 63, 68]). Further development of microbial colourants was a prospective repeatedly mentioned (e.g., [17, 18, 27, 62]). As large-scale dye plant monocultures were not considered a feasible future response to dye demand, natural colourant production was hypothesised to comprise local, diverse, and small-scale solutions complementing one another [16]. Doty et al. [25] speculated whether synthetic fibre use might be reduced, as natural colourants tend to work best on natural fibres. However, due to synthetic fibres' considerable market share, they also saw this property as potentially limiting the use of natural colourants.

It may be a source of insight to try and distinguish whether the fairly cautious ideas about the future of natural colourants identified in the literature more reflect the strong hegemony of synthetic colourants or weak alternatives offered by natural ones. Dator ([22], p. 5) describes a common tendency to see the future as “whatever is happening now, extended and perhaps amplified”, an assumption of continuation that is intuitive to humans but highly misleading. The absence of truly disruptive Horizon-3 visions from the literature reviewed may simply be due to the nature of the data, as the original purpose of the source materials was not to explicitly articulate visions or alternative futures. Having said that, the findings leave room for speculating what those more transformative futures might have looked like or what indeed would be considered a disruptive, transformational change in this context – perhaps a complete replacement of synthetic colourants; a new, imaginative colouration method or colourant source yet to be invented; or ceasing to add colour to textiles altogether.

The findings further raise questions about where, how and by whom images of natural colourant futures are

created. The concept of worldbuilding, borrowed from science fiction to futures research and design, suggests that to create truly different and disruptive new horizons, one needs to systematically imagine plausible and complete worlds for the context of the story at hand, and ask what drives that world [91, 54]. The Horizon-3 visions in our review appear to take place in a world very similar to the one we live in currently. Constrained by the present rules and rationales, the visionary Horizon-3 amounts to little more than a slightly altered version of Horizon-1.

Especially to find solutions to the planetary crisis, an ability to imagine alternatives and the previously unimaginable is required [43]. Futures research widely recognises the role of images of the future in shaping present decisions and societal trajectories (e.g., [7], p. 82, [65], [73]). If these images become collectively shared and reproduced sociotechnical imaginaries, they may gain even more power to steer societies [39]. Because of their underlying power, Bell ([7], p. 82) calls for showing interest in the causes and consequences of our images of the future. According to Miller [56], the potential of images of the future does not lie in their ability to predict the future but to re-perceive the present and what is considered conceivable. What we envision and choose to pursue carries significance, as our chosen Horizon-3 aim will have resources devoted to it, improving it and giving it advantage over any other future [79], p. 21). The alternatives will be left in the margins, such as what happened to natural colourants in the advent of synthetic ones. Not only did the development of natural colourants stall but a lot of already existing knowledge got lost when the interest was almost entirely elsewhere.

Horizon-2 themes highlight tensions in transformations

Horizon-2 theme I: From path dependencies to a visionary attitude

The first Horizon-2 theme discusses the roles of mechanisms and mindsets in transformations. In textile dyeing, like elsewhere, there are past decisions and infrastructures that create path dependencies – technological or social structures that shape our present and future by restricting or channelling our choices (e.g., [82]). Unless rewired, these established structures and production processes keep the fashion industry locked-in and change resistant [13]. As our embedded and often shared worldviews can constrain our capacity to imagine alternative futures, path dependencies in our values and beliefs should also be considered [37].

In the texts reviewed, indications of path-dependent thinking could be found, for example, in the concerns that not enough natural dye sources exist for an unlimited variation of colour choices [92], or that fast-changing trends pose a challenge to working with natural colourants [5]. Infrastructural path dependencies are caused by

existing machinery and processes in dyehouses, often not directly suitable to natural dyes (e.g., [78, 92]. Commercial success of natural colourants was noted to require cheap, large-scale production, which was considered difficult to achieve with natural colourants not necessarily consistently available or offering uniform dyeing results (e.g., [62], [92]. Our established conventions of quality and aesthetics are challenged by the non-permanent and non-uniform nature of natural colourants.

Resisting path dependencies requires cultivating a visionary attitude. This attitude was present in the literature, for example, in references of natural colourants as innovations [6, 18, 59]. Governments play a role in fostering these innovations, and innovation funding should be allocated according to environmental impact, which in textiles points to weaving, dyeing, and finishing stages [79], p. 23, [67]. Uddin and colleagues [85] similarly conclude in their review, that to revive natural colourants, policy support and investments in large-scale production of promising colourants are required. However, innovations were portrayed in the literature not only as technological novelties, but taking place along the value chain, for example in design and marketing [59].

The literature considered design and product development to be important in advancing the use of natural dyes by creating practical and timely styles and products with the characteristics of natural dyes and consumer preferences in mind (e.g., [20, 24, 50, 89]. The visionary orientation in the studies we analysed imagined new aesthetics of uniqueness and imperfection, and a reassessment of how quality in textiles is perceived [59], [60], [46].

Design can play a significant role in reproducing or reimagining what is considered aesthetic, valuable or realisable. The reimagining can be supported by ideas from speculative design, critical design, and neighbouring and interrelated traditions like design fiction and design futures (e.g., [14, 26, 51]. When design is distanced from industrial production and market pressures, more room is created for design to act as a catalyst for visions – it becomes possible to design for how the world could be, not for how it is, and to design to change us to suit the world, not for the world to suit us [26], pp. vii, 9, 12, [51], p. 55). Experimental and unconventional ways to add colour patterns to textiles have been tested by, e.g., Talman [81], who buried textiles underground, immersed them in water, and let signs of use change the textiles' appearance. In doing so, Talman questions permanence as a quality defining sign in textiles and asks if change over time or in different conditions could be embraced. These kinds of design trials are not just to experiment and entertain, but they can critique and challenge us to truly imagine alternatives to the present ways of being [26], p. 34). Nuutinen et al. [61] have also reflected on the challenges of breaking away from the conventions of what is

considered beautiful or ugly, good or bad in design, and the importance of trying to do so nonetheless, to open up to alternative aesthetics and futures.

Particularly visionary and transformative ideas in the literature speculated with textile industry paradigm change. A new industry logic towards slowness, locality and sustainability would offer a paradigm favouring natural colourants [46, 59, 60, 86]. Systems thinker Meadows ([55], pp. 162–163) has described paradigms as “the deepest set of beliefs about how the world works”, and places paradigms high on the list of leverage points for changing systems. According to Meadows ([55], p. 164), paradigms can shift by building a system model that allows us to see the system from the outside and to see it whole. When the system's structures and paradigmatic assumptions are made explicit, opportunities are created to challenge them. This system model of natural textile colourants is complemented by each study adding to the understanding of the colourants, their use, and users. Nurturing a visionary attitude can thus pave the way for a system-level perspective, the second Horizon-2 theme.

Horizon-2 theme II: From fragmentation and disconnection to systemic collaboration and goals

The second theme in Horizon-2 highlights systems perspective in transformations. In the literature reviewed, different emphases on the key determinants of the viability of natural colourants in the market were present. Acceptance and demand; dye production capacity and availability; and dye performance were repeatedly discussed and sometimes raised as concerns (e.g., [50, 62, 68, 78]. The solutions depend on which one of these is considered the primary challenge that needs to be resolved. Gaps in knowledge, experience, information, awareness and access to natural colourants were also recognised to restrain their wider use among all actors in the production and consumption chain (e.g., [4], [25], [50], [59], possibly making natural colourants and natural dyed products unfamiliar and intimidating to work with or use. These notions in the literature disclose the multifaceted and interdependent properties of natural colourants. Fragmentation and disconnection within the industry can happen if parts of the system are not communicating, responsibilities are passed on from one actor to another, or a broader view is lost. The textile industry has a notoriously complex and distributed value chain, which also reflects in natural dye development and adoption.

Deliberately aiming for a systemic view and goals can help navigate these complexities. Many natural colourant related practices endorsed in the literature promote a systemic mindset over siloed thinking: knowledge sharing, transdisciplinary research, open communication, and brave collaboration across natural colourant supply chain and between all stakeholders (e.g., [25, 32, 59, 75].

These efforts materialise, e.g., in education for designers [48], awareness raising for consumers (e.g., [34, 50, 93], technical support for dyers and dyehouses [6, 33], and market understanding and marketing strategy (e.g., [25, 33, 88]. These findings align with Rognoli et al. [72] who suggest that for the development and adoption of bio-based textile materials, communication and marketing strategies are needed to disseminate relevant knowledge efficiently to the public, companies, and designers. Pucker [67] calls for cross-sector collaboration between leading fashion brands to drive the industry's sustainability efforts, and Ketonen-Oksi and Vigren [43] additionally conclude that transformations require collaboration beyond conventional boundaries.

A systemic view acknowledges the multitude of actors and power structures in textile colouration. Consumers' role was accentuated in the literature, not only as the customers for natural dyed products, but also in defining whether natural colourants make it on the market or not [6, 50, 62]. However, excessively highlighting consumers' role may overestimate their abilities to make rational decisions, especially as the information on colourants was often considered to be insufficient [28, 50], misleading, or simply false, as dyes or products marketed as natural dyed are known to have been adulterated [42, 66, 78]. Focusing on one group of actors can also undermine other actors' roles and responsibilities, such as in regulating and maintaining unsustainable production practices. Policymakers particularly are important players, but also those with power to affect other players' actions and change the rules [79], pp. 7–8). Changing the rules and goals can drive system-level change [55], pp. 158, 161) that is not achieved by focusing on single actors or practices. If textile industry system goals remain unchanged, focusing on financial targets, quality, and delivery dates, no room is left for significant environmental improvements to materialise [67]. Identifying the most powerful actors in textile colouration could also reveal if some of that power should be redistributed. For example, dye-houses were sometimes portrayed in the literature as the gatekeepers of what the industry is willing and able to do (e.g., [6]. Meanwhile, fashion brands were reported to show interest in natural colourants [18], [25], [59], [89], but the brands' role in collectively leveraging their potential to drive natural colourant development and implementation was not evident in the literature, beyond a few actively engaged exceptions.

The many sustainability problems surrounding textiles have often been uncovered by parties outside the industry, resulting in reactive and partial responses from both the fashion industry and governments [13]. Developing and implementing natural colourants offers an opportunity for the industry to take proactive responsibility, as large-scale adoption of these colourants requires a

change at the system level, a recalibration of goals and practices. To focus on long-term systemic goals, not short-term minor improvements – to maintain transformational intent – is the first of Fazey and Leicester's [31] six lessons for stewarding transformations. Sustainability as one such systemic goal is the next Horizon-2 theme.

Horizon-2 theme III: From process complexity and poor performance to sustainability benefits and prerequisites

The third and final Horizon-2 theme addresses the choices and trade-offs of transformations. The dye extraction and dyeing processes for natural colourants were described in the literature as complex, time-consuming, laborious, and costly (e.g., [17, 50, 68], and dye performance often questionable, particularly in terms of colour fastness properties (e.g., [18, 62, 78]. We take our reference points for dye production and performance from our experience with synthetic dyes, and for natural colourants that are not necessarily uniform or easily controlled this comparison places unachievable standards (e.g., [25]. The literature notes that this disadvantage could be balanced by new standards and criteria for testing and assessment, where natural colourants' properties would be considered (e.g., [46], another task that could be promoted by regulation.

If still lacking in scale, speed or ease of use, the superior qualities of natural colourants are their renewability, biodegradability and circularity potential, acknowledged in most reviewed studies (see, e.g., [25, 33, 78]. These properties align natural colourants with the circular economy framework which has been prominent in the sustainability discourses of the past two decades, including European Union's initiatives for sustainable textiles (e.g., [40, 44], [30]. Many studies also recognised that some natural colourants have functional advantages, such as antibacterial or UV-resistant properties (e.g., [17, 50, 88]. Potential social sustainability benefits of natural colourants were present in the literature as income creation (e.g., [33, 78] and preservation of local traditions and heritage (e.g., [23, 28, 90]. Colour tourism and colouring workshops for consumers (e.g., [1] are examples of activities that can simultaneously create income, conserve culture and skills, and raise awareness. In South Korea and Indonesia, where natural dyeing has traditional cultural value, the government's active role in supporting it was reported [35], [45].

Balancing the pros and cons of synthetic and natural colourants makes us confront our values. In one of the texts, a clothing brand representative concluded that the challenge with natural colourants is the requirement for commercial products to be perfect [59]. A comprehensive re-evaluation of perfection in the industry is perhaps called for, one that includes other aspects than just visual. Our fondness of and custom to infinite, inexpensive,

and everlasting colours is confronted with the planet's regenerative limits. As each transformation results in new human–nature relations, winners and losers, and trade-offs, asking who gains and who loses, has agency, and decides on the trajectory, rate and scale of change is important [9]. In the textile context this discussion is highly needed.

One option for negotiating the competing interests and values is to firmly keep sustainability benefits and prerequisites as the guiding principles when developing textile colouration practices. Following the strong sustainability view (e.g., [57], pp. 2–3), a sustainable colourant is not a compromise between environmental, social, and economic factors, but exists above all on environmental conditions that are not tradeable for economic benefit. “Creativity energized by boundaries can advance progress”, writes Pucker ([67], p. 33) about sustainability-led design. Actively communicating the sustainability benefits of natural colourants was seen important in the literature, and “telling the story” of the products was considered an asset in distinguishing natural dyed textiles from their synthetic dyed counterparts and justifying their typically higher price (e.g., [15, 38, 49, 50]. Policies can be leveraged to weight some values and goals over others, and the potential power of regulation to support the readoption of natural colourants was repeatedly noted in the literature (e.g., [23, 35, 45, 92]. This potential remains largely underutilised.

Finally, while the papers reviewed touched on the topic of problematic textile production and consumption scale, more attention was given to the insufficiency of natural colourants to currently reach that volume than to questioning the volume itself. As natural colourant production and dyeing practices, too, require resources and have environmental and social impacts, we see that efforts to scale up natural colourants need to be coupled with careful, long-term evaluation and responsible management of these resources and impacts. The uncertainties still surrounding sustainable large-scale use of natural colourants can catalyse imagining what would the H2+ actions look like if we committed to a textile industry that only produced and dyed what was sufficient, feasible and safe inside the planet's life-sustaining capacities.

Reflections, limitations and future research

Alvesson and Sandberg [2] note that while attempting to provide an objective view of a field, literature reviews typically adopt the field's paradigmatic and other assumptions. They recommend reflecting on what may be problematic or constraining in a research community's way of thinking about its domain. We have attempted to do so in our analysis of the literature and invite similar consideration towards our synthesis.

As reflexive thematic analysis embraces researcher subjectivity, and discussing possible, probable and preferred futures is always contested, this review does not claim researcher neutrality. We advocate for a more sustainable textile industry and see natural colourants as potential part of that transformation. While the subjectivity of the analysis method is understood here as a resource rather than a risk [12], we acknowledge it results in limited reproducibility of the analysis and context-specificity of findings. Reflexive thematic analysis shares these characteristics with other qualitative approaches.

The present research design and data carry additional limitations that future research should consider. First, the literature, too, was generally oriented towards exploring the potentials of natural colourants, which is likely a reflection of research motivation within the field – those researching natural colourants tend to see promise in them. Because of this homogeneity of the literature, alternative or critical imaginaries of natural colourants may be underrepresented and underexplored. Second, although a variety of textile industry actors were represented in the datasets of the literature reviewed, the identified imaginaries are constructed by researchers' framings and analyses. Recognising images of the future as drivers of change, we suggest that systematically examining different textile industry stakeholders' imaginaries could illuminate key bottlenecks, trade-offs, and opportunities in sustainability-oriented textile colouration transformation. Third, while giving the less-studied cultural, societal, and applied research the spotlight in this review, we acknowledge that this division to technical versus socio-cultural studies is arbitrary. The prospects of natural colourants could have appeared different in a technology-focused body of literature compared to the present data – a comparison that would make an interesting future research topic. Furthermore, while this review did not differentiate between natural and bio-based colourants, focusing on just one or the other could result in different future projections due to their different production logics.

This review applied the Three Horizons as an analytical framework, and the choice of framework directs attention on certain aspects in the data and takes focus away from others. In this review the Three Horizons provided a simple yet helpful theoretical framing to descriptively explore the present, the near future and the far future dynamics, and for distinguishing in the literature the range of actions and intentions from partial to systemic change. We agree with the framework's particular usefulness as a scoping tool at the start of a project, to frame the problem and identify critical unknowns [80]. As the framework is recognised to depict a simplified, successful transformation, complementing, explanatory approaches are likely needed for deeper insights on scenarios,

transformation trade-offs, as well as system relations and feedbacks [74, 80]. Natural colourants' long history and future prospects also illustrate the framework's heuristic characteristics – in practice, the horizons don't exist and appear neatly and linearly. The past, present and future persist, reappear and interact in layered temporalities.

Applying the framework required interpretative and value judgments on the preferability of visionary futures and on the desirability of stability versus disruption. We have deliberately taken the normative stance where change is considered both necessary and desirable. Consequently, we interpreted as H2– many of the ideas that used Horizon-1 synthetic colourant features as a goal that natural colourants should meet instead of starting off points for something improved. While placing these kinds of requirements for natural colourants can be either constraining or catalytic for their development, we have used the H2–/H2+ lens to challenge which features in colourants are non-negotiable and which may more reflect convention than necessity. Applying the framework to an analysis of existing literature instead of using it in a dialogue-enabling, co-creative context relied primarily on our researcher interpretation and may not fully reflect the reviewed authors' original intentions.

Conclusions

This review has brought together natural colourants and futures perspectives to discuss textile industry sustainability transformation and the performative and instrumental role of images of the future in shaping and informing present actions. Guided by the Three Horizons framework, distinctions were made between thinking that drives incremental change within an existing system and transformative change that aims to redefine the system.

The literature recognised the (hidden) costs of currently dominant synthetic dyes as the primary motivation for finding alternatives to Horizon-1. A growing, if guarded, interest towards Horizon-3 natural colourants from all textile industry stakeholders was consistently present across the studies reviewed. Horizon-2 tensions between a transformational intent and our established ways to 'know and make' textiles were also observed in the literature. To work with these tensions requires the textile industry with its wider ecosystem to consider and address structural and mindset lock-ins, fragmented system perspectives, and the unavoidable trade-offs of change.

The past, present and future coexist in natural colourants that have a long history and renewed future relevance. These entangled temporalities are negotiations between what is imagined, what emerges and what is being replaced, but natural colourants illustrate how transformation can also be about what persists, returns

or is reintroduced. Over the past 170 years synthetic dyes have shaped our understanding and expectations of what it is like to work with and live with textiles. These expectations are met with the planetary limits of today in Horizon-2. If the Horizon-2 decisions favour current manufacturing infrastructure, fast changing trend cycles, and systematically underpriced textiles, Horizon-2 efforts to transition towards natural colourants are likely to result merely as incremental improvements to Horizon-1. If the Horizon-2 is to build conditions for a more sustainable Horizon-3, many dominant paradigmatic assumptions of the textile industry need reframing: quality should be seen not only as technical flawlessness but environmental integrity; economic profit as systemic, long-term viability rather than short-term cost-efficiency; and planetary boundaries to scale and volume as design conditions rather than constraints. These reframings require supporting policies and industry-wide commitment, but they may also need more compelling and articulated Horizon-3 visions that offer clear direction for actions. As an inherently creative, imaginative and future-oriented industry – one that has started a revolution before – the fashion and textile industry undoubtedly has capabilities to imagine futures that are meaningfully different from the present.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40309-026-00290-y>.

Supplementary Material 1.

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Authors' contributions

LV, PF and RR conceptualised and designed the study. LV conducted the literature search, synthesised the findings, and drafted the first version of the manuscript. PF and RR revised the manuscript and supervised the process. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available in published sources cited in Table 1 and in this article's reference list.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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References

- Agustini MH, Athanasius SS, Retnawati BB (2019) Identification of green marketing strategies: perspective of a developing country. *Innov Mark* 15(4):42–56. [https://doi.org/10.21511/im.15\(4\).2019.04](https://doi.org/10.21511/im.15(4).2019.04)
- Alvesson M, Sandberg J (2020) The problematizing review: a counterpoint to Elsbach and van Knippenberg's argument for integrative reviews. *J Manag Stud* 57(6):1290–1304. <https://doi.org/10.1111/joms.12582>
- ATLAS.ti Scientific Software Development GmbH (2025) ATLAS.ti (version 25.0.1.32924) [Qualitative data analysis software]. <https://atlasti.com>
- Autio M, Poikolainen J, Sekki S, Ruippo L, Räisänen R (2023) Consumers' concern about chemicals in clothing – environment, skin contact, and the materiality of colour. In: Niinimäki K, Cura K (eds) 5th PLATE Conference Proceedings, Aalto University publication series, pp 71–76
- Bechtold T, Manian A (2023) Technical aspects and requirements to produce natural colorants – processes and product standardization. In: Bechtold T, Manian AP, Pham T (eds) Handbook of natural colourants, 2nd ed. Wiley, Hoboken, pp 573–589. <https://doi.org/10.1002/9781119811749.ch28>
- Bechtold T, Pham T, Manian A (2023) Textile coloration with natural dyes and pigments. In: Bechtold T, Manian AP, Pham T (eds) Handbook of natural colourants, 2nd ed. Wiley, Hoboken, pp 517–541. <https://doi.org/10.1002/9781119811749.ch24>
- Bell W (2009) Foundations of futures studies. Volume 1: history, purposes, and knowledge. Transaction Publishers, New Brunswick
- Bentz J, O'Brien K, Scoville-Simonds M (2022) Beyond “blah blah blah”: exploring the “how” of transformation. *Sustain Sci* 17(2):497–506. <https://doi.org/10.1007/s11625-022-01123-0>
- Blythe J, Silver J, Evans L, Armitage D, Bennett NJ, Moore ML, Morrison TH, Brown K (2018) The dark side of transformation: latent risks in contemporary sustainability discourse. *Antipode* 50(5):1206–1223. <https://doi.org/10.1111/anti.12405>
- Braun V, Clarke V (2006) Using thematic analysis in psychology. *Qual Res Psychol* 3(2):77–101. <https://doi.org/10.1191/1478088706qp0630a>
- Braun V, Clarke V (2022) Thematic analysis: a practical guide. SAGE Publications, London
- Braun V, Clarke V (2023) Toward good practice in thematic analysis: avoiding common problems and becoming a *knowing* researcher. *Int J Transgend Health* 24(1):1–6. <https://doi.org/10.1080/26895269.2022.2129597>
- Buchel S, Hebinck A, Lavanga M, Loorbach D (2022) Disrupting the status quo: a sustainability transitions analysis of the fashion system. *Sustain Sci Pract Policy* 18(1):231–246. <https://doi.org/10.1080/15487733.2022.2040231>
- Candy S, Kornet K (2019) Turning foresight inside out: an introduction to ethnographic experiential futures. *J Futures Stud* 23(3):3–22. [https://doi.org/10.6531/JFS.201903_23\(3\).0002](https://doi.org/10.6531/JFS.201903_23(3).0002)
- Cao H, Scudder C, Howard C, Piro K, Tattersall H, Frett J (2014) Locally produced textiles: product development and evaluation of consumers' acceptance. *Int J Fash Des Technol Educ* 7(3):189–197. <https://doi.org/10.1080/17543266.2014.950347>
- Cardon D (2023) Colour cultures: natural dyes, a global intangible heritage with a future. *Acta Hort* 1361:1–12. <https://doi.org/10.17660/ActaHortic.2023.1361.1>
- Carvalho C, Santos G (2015) Global communities, biotechnology and sustainable design – natural/bio dyes in textiles. *Procedia Manuf* 3:6557–6564. <https://doi.org/10.1016/j.promfg.2015.07.956>
- Casciani D, Chen M (2024) Conveying natural dyes in the fashion industry through a design-driven innovation. In: Cantista I, Ritch EL, Shearer L, Pérez-Bou S, Khar SS (eds) Fashion for the common good, GFC 2023. Palgrave Macmillan, Cham, pp 46–64. https://doi.org/10.1007/978-3-031-50252-1_4
- Child M, Breyer C (2017) Transition and transformation: a review of the concept of change in the progress towards future sustainable energy systems. *Energy Policy* 107:11–26. <https://doi.org/10.1016/j.enpol.2017.04.022>
- Choi J, Kim S (2013) Cross-cultural study of color sensation and sensibility for Korean traditional fabrics with natural indigo dyeing. *Fibers Polym* 14(1):138–145. <https://doi.org/10.1007/s12221-013-0138-z>
- Curry A, Hodgson A (2008) Seeing in multiple horizons: connecting futures to strategy. *J Futures Stud* 13(1):1–20
- Dator J (2009) Alternative futures at the Manoa School. *J Futures Stud* 14(2):1–18
- de la Sayette A (2023) Large-scale production of dye plants and plant extracts for industrial applications of natural dyes. *Acta Hort* 1361:131–140. <https://doi.org/10.17660/ActaHortic.2023.1361.15>
- Dong ZJ, Liang JF, Zhang ZJ, Wei SS (2023) The perceptual evaluation of clothing sustainable color in clothing design. *J Fiber Bioeng Inform* 16(3):229–241. <https://doi.org/10.3993/jfbim02351>
- Doty K, Green DN, Haar S (2024) Natural dyes in the United States: an exploration of natural dye use through the lens of the circuit of style-fashion-dress. *Cloth Text Res J* 42(4):310–324. <https://doi.org/10.1177/0887302X241229222>
- Dunne A, Raby F (2013) Speculative everything: design, fiction and social dreaming. The MIT Press, Massachusetts
- Elsahida K, Fauzi AM, Sailah I, Siregar IZ (2019) Sustainability of the use of natural dyes in the textile industry. *IOP Conf Ser Earth Environ Sci* 399:012065. <https://doi.org/10.1088/1755-1315/399/1/0120652036>
- Elsahida K, Fauzi AM, Sailah I, Siregar IZ (2020) Sustainable production of natural textile dyes industry. *IOP Conf Ser Earth Environ Sci* 472:012036. <https://doi.org/10.1088/1755-1315/472/1/01>
- Elsbach KD, van Knippenberg D (2020) Creating high-impact literature reviews: an argument for 'integrative reviews.' *J Manag Stud* 57(6):1277–1289. <https://doi.org/10.1111/joms.12581>
- European Commission (2022) EU Strategy for Sustainable and Circular Textiles (COM(2022) 141 final). Publications Office of the European Union, Brussels. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52022DC0141>
- Fazey I, Leicester G (2022) Archetypes of system transition and transformation: six lessons for stewarding change. *Energy Res Soc Sci* 91:102646. <https://doi.org/10.1016/j.erss.2022.102646>
- Ferrante M, Arzoumanidis I, Petti L (2019) Socio-economic effects in the knitwear sector – a life cycle-based approach towards the definition of social indicators. In: Muthu SS (ed) Social life cycle assessment. Environmental footprints and eco-design of products and processes. Springer, Singapore, pp 59–97. https://doi.org/10.1007/978-981-13-3233-3_2
- Geissler S, Bechtold T (2023) Environmental and economic position of natural colorants – energy and resources balances, sustainability, ecology, and costs. In: Bechtold T, Manian AP, Pham T (eds) Handbook of natural colourants, 2nd ed. Wiley, Hoboken, pp 591–612. <https://doi.org/10.1002/9781119811749.ch29>
- Hashim NIR, Md Salleh NA, Norazlimi NA, Rahman AAA, Pa'ee F (2021) A preliminary study of tourists and vendors perception on the development of tourism souvenirs using natural dye from plants. *IOP Conf Ser Earth Environ Sci* 736:012020. <https://doi.org/10.1088/1755-1315/736/1/012020>
- Hong H, Kang JH (2019) The impact of moral philosophy and moral intensity on purchase behavior toward sustainable textile and apparel products. *Fash Text* 6:16. <https://doi.org/10.1186/s40691-019-0170-8>
- Hölscher K, Wittmayer JM, Loorbach D (2018) Transition versus transformation: what's the difference? *Environ Innov Soc Trans* 27:1–3. <https://doi.org/10.1016/j.jeist.2017.10.007>
- Inayatullah S (1998) Causal layered analysis: poststructuralism as method. *Futures* 30(8):815–829. [https://doi.org/10.1016/S0016-3287\(98\)00086-X](https://doi.org/10.1016/S0016-3287(98)00086-X)
- Ingram W (2019) Deities, dyers, and dollars: balancing culture, conservation, and commerce with Indonesia's indigenous weavers. *Textile* 17(2):120–134. <https://doi.org/10.1080/14759756.2018.1473982>
- Janoff S (2015) Future imperfect: science, technology, and the imaginations of modernity. In: Janoff S, Kim S-H (eds) Dreamscapes of modernity: socio-technical imaginaries and the fabrication of power. University of Chicago Press, Chicago, pp 1–33. <https://doi.org/10.7208/chicago/9780226276663.003.0001>
- Jia F, Yin S, Chen L, Chen X (2020) The circular economy in the textile and apparel industry: a systematic literature review. *J Clean Prod* 259:120728. <https://doi.org/10.1016/j.jclepro.2020.120728>
- Jordan J (2021) The future of unmanned combat aerial vehicles: an analysis using the Three Horizons framework. *Futures* 134:102848. <https://doi.org/10.1016/j.futures.2021.102848>
- Karadag R (2023) Establishing a new international standard for natural dyed textile goods [Natural Organic Dye Standard (NODS)]. *J Nat Fibers* 20(1):2162187. <https://doi.org/10.1080/15440478.2022.2162187>
- Ketonen-Oksi S, Vigren M (2024) Methods to imagine transformative futures. An integrative literature review. *Futures* 157:103341. <https://doi.org/10.1016/j.futures.2024.103341>

44. Kirchherr J, Yang N-HN, Schulze-Spüntrup F, Heerink MJ, Hartley K (2023) Conceptualizing the circular economy (revisited): an analysis of 221 definitions. *Resour Conserv Recycl* 194:107001. <https://doi.org/10.1016/j.resconrec.2023.107001>
45. Kusumastuti A, Atika ATA, Phusavat K, Hidayanto AN (2023) Assessment of producer's perspective on the production of environmentally friendly fashion products: a case study in Indonesian natural dyes batik craftsmen. *Environ Sci Pollut Res* 30:124767–124779. <https://doi.org/10.1007/s11356-022-23330-z>
46. Ladekarl IM, Harsaae M, Bang AL, Kramer Pedersen L, Bull Nielsen TC, Ege A (2023) From craft to industry. *FormAkademisk* 16(4). <https://doi.org/10.7577/formakademisk.5478>
47. Lai C-C, Chang C-E (2021) A study on sustainable design for indigo dyeing color in the visual aspect of clothing. *Sustainability* 13(7):3686. <https://doi.org/10.3390/su13073686>
48. Lara L, Cabral I, Cunha J (2023) Color in sustainable fashion: a reflection on the importance of design education. In: Raposo D, Neves J, Silva R, Correia Castilho L, Dias R (eds) *Advances in design, music and arts II*. EIMAD 2022. Springer, Cham, pp 238–247. https://doi.org/10.1007/978-3-031-09659-4_18
49. Lin PMC, Peng K-L, Ren L (2015) Xinpū persimmon dye: evolution from a local to a global industry. *J China Tour Res* 11(2):214–228. <https://doi.org/10.1080/19388160.2015.1043066>
50. Mabuza L, Sonnenberg N, Marx-Pienaar N (2023) Natural versus synthetic dyes: consumers' understanding of apparel coloration and their willingness to adopt sustainable alternatives. *Resour Conserv Recycl Adv* 18:200146. <https://doi.org/10.1016/j.rccadv.2023.200146>
51. Malpass M (2017) *Critical design in context. History, theory and practices*. Bloomsbury, London
52. Mamun MAA, Haji A, Mahmud MH, Repon MR, Islam MT (2023) Bibliometric evidence on the trend and future direction of the research on textile coloration with natural sources. *Coatings* 13(2):413. <https://doi.org/10.3390/coatings13020413>
53. Market Growth Reports (2026) *Dyes and pigments market size, share, growth, and industry analysis, by type (dyes, organic pigments, inorganic pigments), by application (textiles, leather, paper, paints & coatings, plastics, constructions, paper & specialty, printing inks, others), regional insights and forecast to 2034*. Report No. 117561. <https://www.marketgrowthreports.com/market-reports/dyes-and-pigments-market-117561> (accessed on 3.2.2026)
54. McDowell A (2019) Storytelling shapes the future. *J Futures Stud* 23(3):105–112. [https://doi.org/10.6531/JFS.201903_23\(3\).0009](https://doi.org/10.6531/JFS.201903_23(3).0009)
55. Meadows D (2008) *Thinking in Systems. A Primer*. In: Wright D (ed). Chelsea Green Publishing, White River Junction
56. Miller R (2010) *Futures literacy - embracing complexity and using the future*. *Ethos* 10(10):23–28
57. Neumayer E (2025) *Weak versus strong sustainability. Exploring the limits of two opposing paradigms*, 5th ed. Edward Elgar Publishing, Cheltenham
58. Niinimäki K, Peters G, Dahlbo H, Perry P, Rissanen T, Gwilt A (2020) The environmental price of fast fashion. *Nat Rev Earth Environ* 1(4):189–200. <https://doi.org/10.1038/s43017-020-0039-9>
59. Niinimäki K, Tommila N, Räsänen R (2023) Colour collaboration. In: Niinimäki K, Lohmann J (eds) *Biocolours: sustainable stories from nature, lab and industry*. Aalto University, Espoo, pp 293–311
60. Niinimäki K, Uribe LH (2023) Sustainable colour aesthetics. In: Niinimäki K, Lohmann J (eds) *Biocolours: sustainable stories from nature, lab and industry*. Aalto University, Espoo, pp 217–233
61. Nuutinen A, Räsänen R, Fernström P (2017) Ugly as a concept in craft to examine alternative futures. *Eur J Futures Res* 5:18. <https://doi.org/10.1007/s40309-017-0127-3>
62. Panda A, Maiti S, Madiwale P, Adivarekar R (2022) Natural dyes – a way forward. In: Pandit P, Singha K, Maity S, Ahmed S (eds) *Textile dyes and pigments: a green chemistry approach*. Scrivener Publishing, Beverly, pp 323–343. <https://doi.org/10.1002/9781119905332.ch16>
63. Patel BH (2011) Natural dyes. In: Clarke M (ed) *Handbook of textile and industrial dyeing. Principles, processes and types of dyes*. Woodhead Publishing, Cambridge, pp 395–424. <https://doi.org/10.1533/9780857093974.2.395>
64. Planetary Boundaries Science (2025). *Planetary Health Check 2025*. Kitzmann NH, Caesar L, Sakschewski B and Rockström J (eds). <https://doi.org/10.48485/pik.2025.017>
65. Polak F (1973) *The image of the future*. Elsevier, Amsterdam
66. Primetta A, Räsänen R (2023) New biocolorant database for the quality control of natural colorants and products containing them. *Acta Horti* 1361:189–194. <https://doi.org/10.17660/ActaHort.2023.1361.22>
67. Pucker K (2023) A circle that isn't easily squared. *Stanf Soc Innov Rev* 21(3):26–39. <https://doi.org/10.48558/96N0-Q813>
68. Ratnapandian S (2020) Natural colorants and its recent developments. In: Nayak R (ed) *Sustainable technologies for fashion and textiles*. Woodhead Publishing, Cambridge, pp 189–208. <https://doi.org/10.1016/B978-0-08-102867-4.00009-8>
69. Raworth K (2017) *Doughnut economics: seven ways to think like a 21st-century economist*. Chelsea Green Publishing, White River Junction
70. Richardson K, Steffen W, Lucht W, Bendsten J, Cornell SE, Donges JF et al (2023) Earth beyond six of nine planetary boundaries. *Sci Adv* 9(37):eadh2458. <https://doi.org/10.1126/sciadv.adh2458>
71. Rockström J, Steffen W, Noone K, Persson Å, Chapin FS III, Lambin EF et al (2009) A safe operating space for humanity. *Nature* 461:72–475. <https://doi.org/10.1038/461472a>
72. Rognoli V, Petreca B, Pollini B, Saito C (2022) Materials biography as a tool for designers' exploration of bio-based and bio-fabricated materials for the sustainable fashion industry. *Sustain Sci Pract Policy* 18(1):749–772. <https://doi.org/10.1080/15487733.2022.2124740>
73. Rubin A (2013) Hidden, inconsistent, and influential: images of the future in changing times. *Futures* 45:38–44. <https://doi.org/10.1016/j.futures.2012.11.011>
74. Rutting L, Vervoort J, Mees H, Pereira L, Veeger M, Muiderman K, Mangnus A, Winkler K, Olsson P, Hichert T, Lane R, Bottega Pergher B, Christiaens L, Bansal N, Hendriks A, Driessen P (2023) Disruptive seeds: a scenario approach to explore power shifts in sustainability transformations. *Sustain Sci* 18:1117–1133. <https://doi.org/10.1007/s11625-022-01251-7>
75. Räsänen R (2023) Networking in biocolour research – creating increasing value and impact. *Acta Horti* 1361:173–180. <https://doi.org/10.17660/ActaHort.2023.1361.20>
76. Sahimaa O, Miller EM, Halme M, Niinimäki K, Tanner H, Mäkelä M, Rissanen M, Härrä A, Hummel M (2024) From simplistic to systemic sustainability in the textile and fashion industry. *Circ Econ Sust* 4:1115–1131. <https://doi.org/10.1007/s43615-023-00322-w>
77. Saldaña J (2021) *The coding manual for qualitative researchers*, 4th ed. SAGE Publications, Thousand Oaks
78. Saxena S, Raja ASM (2014) Natural dyes: sources, chemistry, application and sustainability issues. In: Muthu SS (ed) *Roadmap to sustainable textiles and clothing. Textile science and clothing technology*. Springer, Singapore, pp 37–80. https://doi.org/10.1007/978-981-287-065-0_2
79. Sharpe B (2020) *Three horizons: the patterning of hope*, 2nd ed. Triarchy Press, Axminster
80. Sharpe B, Hodgson A, Leicester G, Lyon A, Fazey I (2016) Three horizons: a pathways practice for transformation. *Ecol Soc* 21(2):47. <https://doi.org/10.5751/ES-08388-210247>
81. Talman R (2018) Designing for multiple expressions: questioning permanence as a sign of quality in textiles. *J Text Des Res Pract* 6(2):201–221. <https://doi.org/10.1080/20511787.2018.1514697>
82. Tiberius V (2011) Towards a "planned path emergence" view on future genesis. *J Futures Stud* 15(4):9–24
83. Torraci RJ (2016) Writing integrative literature reviews: using the past and present to explore the future. *Hum Resour Dev Rev* 15(4):404–428. <https://doi.org/10.1177/1534484316671606>
84. Travis AS (1993) *The rainbow makers. The origins of the synthetic dyestuffs industry in Western Europe*. Lehigh University Press, Bethlehem
85. Uddin MA, Rahman MM, Haque ANMA, Smriti SA, Datta E, Farzana N, Chowdhury S, Haider J, Sayem ASM (2022) Textile colouration with natural colorants: a review. *J Clean Prod* 349:131489. <https://doi.org/10.1016/j.jclepro.2022.131489>
86. Uribe LH (2023) Making colour a relational practice: correspondence between materials and local environments in natural textile dyeing processes. In: Niinimäki K, Lohmann J (eds) *Biocolours: sustainable stories from nature, lab and industry*. Aalto University, Espoo, pp 107–130
87. Valle-Noronha J, Puppert P (2024) Estonian blues: natural dyes as enhancers of environmental, intergenerational and heritage bonds in fashion design education. *Craft Res* 15(1):41–60. https://doi.org/10.1386/crrc_00118_1
88. Xu H, Song H-y (2024) Key factors influencing Chinese consumers' demand for naturally dyed garments: data analysis through KJ method and KANO model. *Sustainability* 16(3):1189. <https://doi.org/10.3390/su16031189>
89. Xue X, Li L (2023) Fashion consumption of naturally dyed products: a cross-cultural study of the consumption of blue-dyed apparel between China and Japan. *Fib Text East Eur* 31(1):52–65. <https://doi.org/10.2478/ftce-2023-0006>

90. Yi E, Shamey R (2015) Crosscultural comparison of color terms and preference of persimmon-dyed fabric, Galchon. *Color Res Appl* 40(6):592–604. <https://doi.org/10.1002/col.21941>
91. Zaidi L (2019) Worldbuilding in science fiction, foresight and design. *J Futures Stud* 23(4):15–26. [https://doi.org/10.6531/JFS.201906_23\(4\).0003](https://doi.org/10.6531/JFS.201906_23(4).0003)
92. Zerín I, Farzana N, Sayem ASM, Anang DM, Haider J (2020) Potentials of natural dyes for textile applications. In: Hashmi S, Choudhury IA (eds) *Encyclopedia of renewable and sustainable materials*, vol Vol. 2. Elsevier, Amsterdam, pp 873–883. <https://doi.org/10.1016/B978-0-12-803581-8.11668-6>
93. Zhou Y, Falk CL, Vanleeuwen DM (2012) Retail demand for natural dye plants and dye plant products: a conjoint analysis. *J Int Food Agribus Mark* 24(1):66–75. <https://doi.org/10.1080/08974438.2012.645740>

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